

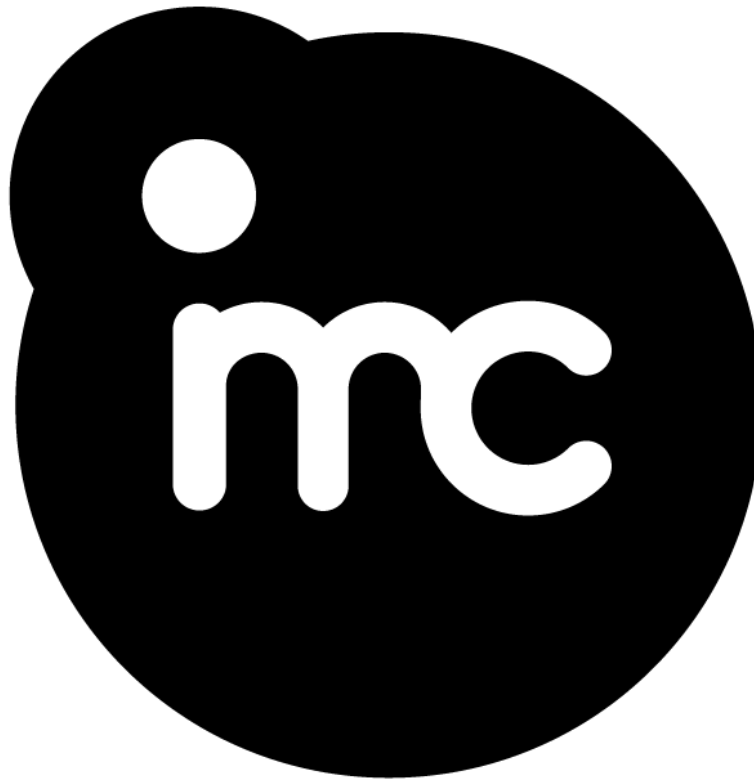


IMC University of Applied Sciences Krems

Title: Documentation for Assignment 2

Course: WS25/26 Informatics / Web Technologies

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Date: 11.12.2025



1. Project Overview:

Hope Supplies is a **Single Page Application** (SPA) designed to facilitate charitable donations. The user interface allows donors to browse a gallery of essential items (such as winter coats, food, and medical supplies), add them to a shopping cart, and proceed to a secure checkout form. The project utilizes Bootstrap 5 for a responsive, mobile-friendly design and JavaScript to handle the application logic, including view navigation, cart management, form validation, and dynamic price calculations without reloading the page.

File Structure:

```
+---2025-webtech-assignment-2-hsynrsd
| | .gitignore
| | Assignment2-JavaScript-Bootstrap.pdf
| | Documentation.pdf
| | index.html
| | README.md
| |
| +---.github
| | \---workflows
| |     classroom.yml
| |
| +---css
| |     style.css
| |
| +---img
| |     backpack.jpg
| |     canned_foods.jpg
| |     first_aid_kit.jpg
| |     formula.jpg
| |     hygiene_kit.jpg
| |     solar_lamp.jpg
| |     tent.png
| |     water_filter.jpg
| |     winter_coat.jpg
| |     wool_blanket.jpg
| |
| +---js
| |     scripts.js
| |
| \---tests
|     test_smoke.py
```

2. Bootstrap Implementation:

2.1. Grid System & Layout:

A responsive grid was used for the product gallery.

- Class: row row-cols-1 row-cols-md-3 row-cols-lg-4 g-4
- Function: This ensures the layout displays 1 column on mobile devices, 3 on tablets and 4 on large desktop screens, providing an optimal user experience across devices.

2.2. Components Used:

I implemented more than the required five Bootstrap components:

- **Navbar** (<nav>): Used for the header, featuring a responsive collapse button for mobile views, it also houses the “Cart” button.
- **Cards** (.card): Used to display the 10 donation items. Each card contains an image, title, description, price and an “Add to Cart” button.
- **Modal** (.modal): Used for the **Shopping Cart**. When the user clicks the “Cart” button in the navbar, a modal overlay appears showing the selected items, allowing the user to review their donation without leaving the gallery.
- **Badges** (.badge): A red notification badge is positioned on the Cart button to display the real-time counts of items added (id=“cart-count”).
- **Forms** (.form-control, .input-group): The checkout section utilizes Bootstrap form styling for inputs labels, and validation states. I also used an Input Group to attach the country code selector to the phone number input.
- **Alerts** (.alert):
 - Info Alert: Displayed in the gallery to inform users about the discount rule.
 - Success Alert: Displayed on the confirmation page to thank the user.

3. Custom CSS Implementation:

While Bootstrap 5 provided the foundation for the grid and components, I utilized a custom style.css file to handle specific design requirements, fix layout consistency issues, and improve user experience.

3.1. Gallery Image Consistency:

- One of the main challenges encountered was that the source images had varying aspect ratios, which caused the Bootstrap cards to be of uneven heights, breaking the grid layout.
- To solve this, I applied a fixed height and the object-fit property to the card images. This ensures all images are exactly 300px tall and crop neatly without distorting (squishing) the picture.

`/* Fix for varying image sizes */`

```
.card-img-top {  
  height: 300px;  
  width: 100%;  
  object-fit: cover; /* Crops image to fit container */  
  border-bottom: 1px solid #ddd;  
}
```

3.2. Visual Interactivity & Polish:

To make the application feel more responsive and modern, I added custom interactions:

- **Card Hover Effect:** Using `transform: scale(1.02)`, the product cards slightly "zoom in" and increase their shadow when the user hovers over them. This provides visual feedback that the item is clickable.
- **Receipt Styling:** For the confirmation page, I styled the `.card` with a 2px dashed `#198754` border and a light green background. This mimics the visual appearance of a physical paper receipt.

3.3. Sticky Footer:

To ensure the footer always stays at the bottom of the viewport (even if the page content is short, such as on the confirmation screen), I applied Flexbox properties to the body and main tags:

```
body {  
    min-height: 100vh;  
    display: flex;  
    flex-direction: column;  
}  
  
main {  
    flex: 1; /* Pushes footer to the bottom */  
}
```

4. JavaScript Logic

4.1. Single Page Application (View Management)

As requested by the requirements of having a single .html file, I used JavaScript to toggle the visibility of specific HTML sections.

- Method: `document.getElementById('section-id').style.display = "none" / "block"`
- Flow: When `proceedToCheckOut()` is called, the Gallery section is hidden, and the checkout form is displayed. Conversely, `goBackToGallery()` reverses this.

4.2. Shopping Cart Logic

- Data Structure: A global array `var cart = []` stores item objects. (I know it's bad practice to do this.)
- Adding Items: The `addToCart(name, price)` function pushes a new object `{name: name, price:price}` into the array and updates the navbar badge count.
- Modal Update: When the modal is opened, the `updateModalContent()` function loops through the cart array and dynamically creates list items (`<li>`) to display the current selection.

4.3. Form Validation

Custom validation was implemented in `handleFormSubmit(e)`:

- Phone Number: A regex (`/^\+?[0-9\s]+\$/`) validates that the phone number contains only digits, spaces, or a plus sign. This prevents invalid characters while allowing international formats.
- ZIP Code: A length check ensures the ZIP Code is not longer than 6 characters. (`zip.length > 6`)
- Error Handling: If validation fails, error messages (`.text-danger`) are displayed immediately below the specific input fields.

4.4. Calculations & Discount

- Subtotal: Sum of all items in the cart.
- Tax: Calculated as 10% of the subtotal.
- Discount: An if statement checks the cart size:

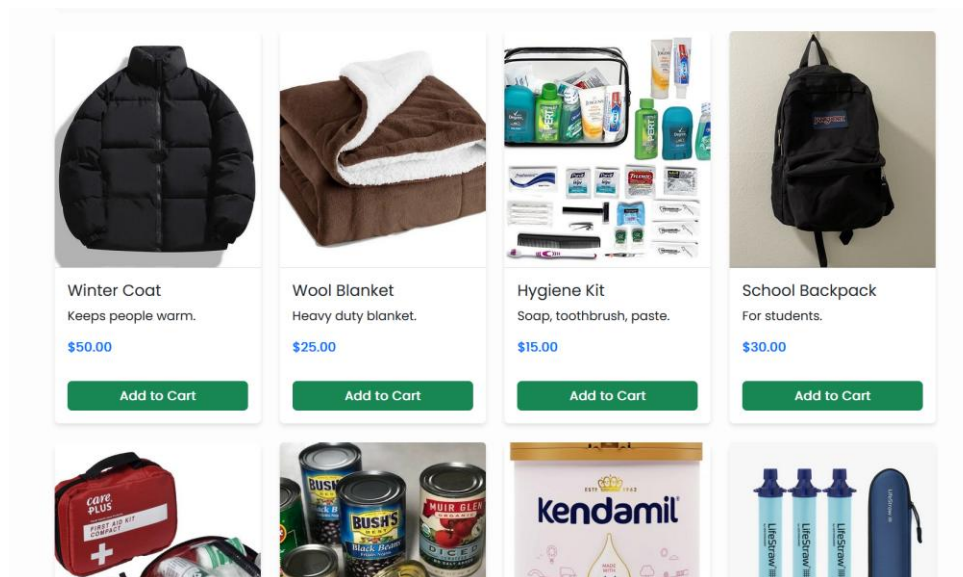
Code:

```
if (cart.length >= 3) {
    discount = 10; // flat 10€ discount.
}
```

This applies a 10€ discount if we have 3 or more items.

5. Visual Walkthrough

5.1. Product Gallery – The initial view displaying the grid of donation items.





Winter Coat
Keeps people warm.
\$50.00

Add to Cart



5.2. Shopping Cart (modal) – The modal view showing items added to the cart.

Shopping Cart

Winter Coat - \$50

Wool Blanket - \$25

Total: \$75

Close

Checkout

5.3. Checkout form (with validation) – The shipping information form. Note the country code selector.



[← Back to Gallery](#)

Shipping Information

Full Name

Hyusein Rashid

Email address

25imc10140@fh-krems.ac.at

Phone Number

Austria (+43) [▼] +43 676123123123

Address

IMC

ZIP Code (Max 6)

3500

City

Krems

Complete Checkout

Full Name

Please fill out this field.

Email address

Hi

Please enter an email address.

Address

Please fill out this field.

City

Phone Number

Austria (+43) [▼] +43 6764602143ads

Invalid phone number (allow numbers, spaces, and +)



ZIP Code (Max 6)

3500000000

ZIP Code too long!

City

Please fill out this field.

[← Back to Gallery](#)

Shipping Information

Full Name

Hyusein Rashid

Email address

25imc10140@fh-krems.ac.at

Phone Number

Austria (+43) +43 676123123123

Address

IMC

ZIP Code (Max 6)

3500

City

Krems

Complete Checkout

5.4. Order confirmation – The final receipt showing the calculated subtotal, tax, and discount.



Donation Successful!

Thank you for your support.

Receipt

Winter Coat - \$50

Wool Blanket - \$25

Subtotal: \$75

Taxes (10%): \$7.50

Discount: -\$0

Total: \$82.50

Start Over



Donation Successful!

Thank you for your support.

Receipt

Winter Coat - \$50

Wool Blanket - \$25

Subtotal: \$75

Taxes (10%): \$7.50

Discount: -\$0

Total: \$82.50

Start Over

6. References

6.1. Bootstrap 5 Implementation: Used for component syntax (Navbar, Cards, Modals, Forms)

- <https://getbootstrap.com/docs/5.3/>

6.2. StackOverflow: Consulted for JavaScript Regex patterns regarding phone number validation (allowing spaces and plus signs).

- <https://stackoverflow.com/questions/16699007/regular-expression-to-match-standard-10-digit-phone-number>

6.3. Pinterest: Source for placeholder product images. I don't think I can find them anymore.

6.4. Remove.bg: Used to create the transparent background for the "Tent": image, that WAS taken from Pinterest too!